



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Secure AI Workflow for Confidential Research

Authorized evidence, controlled tools, and accountable collaboration

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Research Problem

How can researchers use AI and scientific software without exposing confidential data?

Keep data inside

Files and model context stay inside the protected network.

Use permitted evidence

Only approved evidence reaches the model.

Control every action

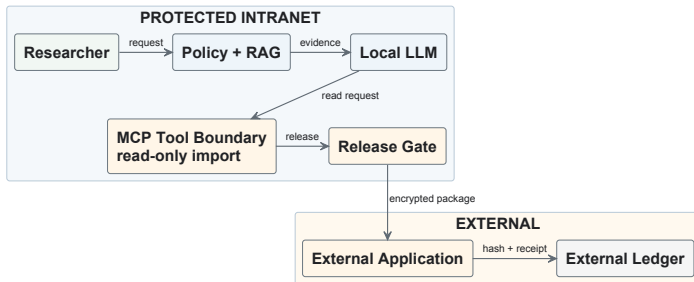
Narrow interfaces control tools and release.

One researcher experience; several independent trust boundaries.

Target Workflow

MCP is the controlled bridge between local AI, approved scientific software exports, and controlled external release.

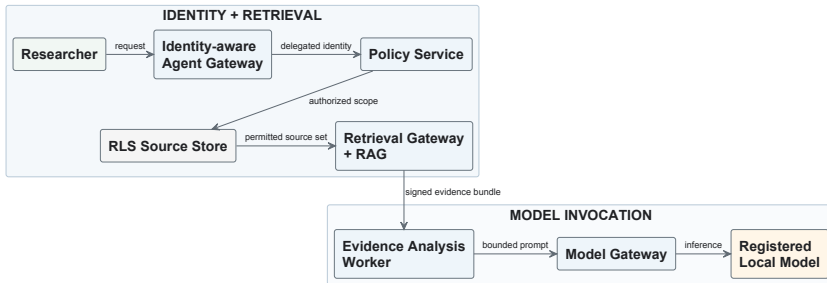
TARGET DESIGN



Authorize Before Retrieval

An identity-aware agent gateway and policy authorize evidence before RAG; a separate model gateway invokes the registered local model.

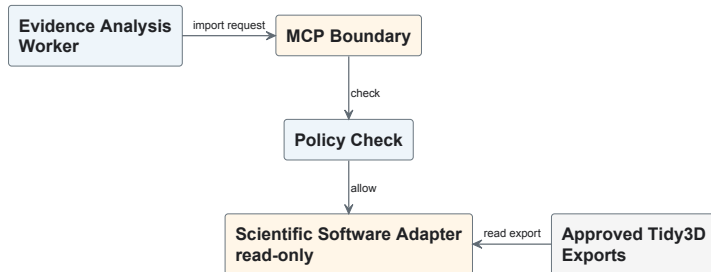
TARGET DESIGN



MCP as the Tool Boundary

The model may request a bounded read-only import through a scientific software adapter; Tidy3D is the initial validation case.

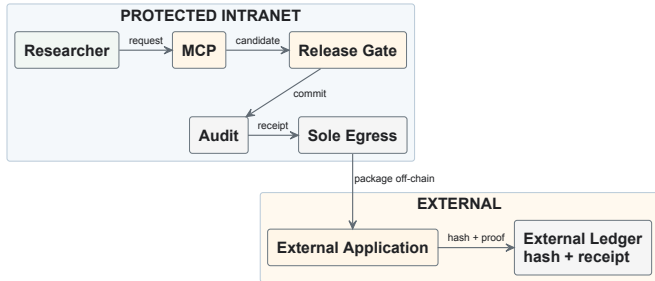
PROPOSED; INITIAL CASE: TIDY3D



Confidential Collaboration

Confidential payloads stay in protected off-chain packages; the ledger stores only a minimal proof.

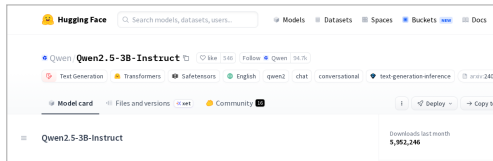
CONCEPT; INTERFACE UNDEFINED



Model Options

Model scale changes deployment; it does not replace authorization or review.

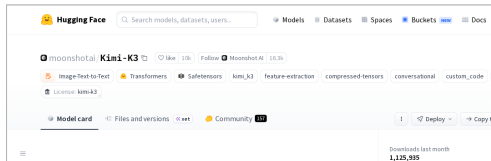
LOCAL-SCALE CANDIDATE



Qwen2.5-3B-Instruct

3B smoke-tested on one synthetic photonics fixture;
not scientific validation.

CLUSTER-SCALE CANDIDATE



Kimi K3

Open-weight; 2.8T / 104B active; cluster-only.

CURRENT EVIDENCE

Current smoke test: qwen2.5:3b via loopback Ollama; qwen2.5:7b remains the handoff baseline.
Neither candidate is enterprise integrated.



What Runs Today

Only the local evidence workflow is operational; the intranet, MCP, and blockchain paths remain future work.

RUNS NOW

Files → traceable evidence → local LLM → checked report

Single user, local access; public or synthetic data.

UNDER REVIEW

Stable interface between the agent and local model

NOT BUILT

**Authorized RAG
Secure release**

**MCP tools
External ledger**



Engineering Validation Path

Validate each trust boundary before connecting scientific tools or external networks.

01 NOW

Local model interface

Finish review; record which model produced each output.



02 NEXT

Authorized retrieval

Test access with synthetic data.



03 THEN

Release protocol

Define approval, encryption, and receipts.



04 GATED

Science tool adapter

Connect one read-only local adapter.



05 LAST

Ledger checkpoint

Use a ledger proof only if a real need remains.

Seven-Day Usage and Cost

01–07 Aug: usage is cache-heavy; a zero price estimate is not a zero operating cost.

LOCAL CC-SWITCH

4,635

requests

597M

total tokens

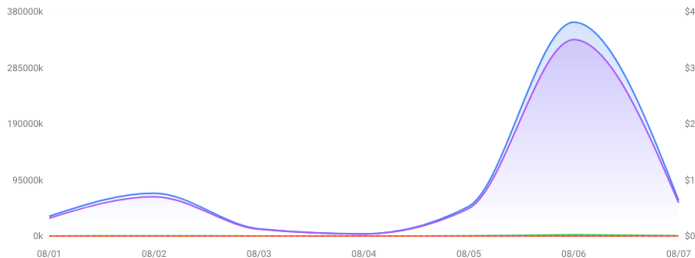
547M

cache reads

\$0.00

interface estimate

Daily token trend from cc-switch



Blue = input tokens purple = cache reads red = estimated cost

Cost interpretation: the \$0.00 interface estimate does not include hardware, cluster, or solver costs.